

MUSCLE GUARD

RESEARCH LIBRARY · COMPREHENSIVE EDITION

The research, in full.

14 plain-English articles on GLP-1 medications,
muscle loss, protein, and resistance training.

With sources you can verify yourself.

Compiled by the Muscle Guard research team.

Last updated 6 June 2026 · Free to share.

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Compiled and last reviewed.

Compiled 6 June 2026 by the Muscle Guard research team. We update this PDF when the underlying science updates. Email research@muscleguardglp.com to flag corrections.

A note on what is — and is not — directly cited.

The un-intervened lean-mass-loss share figure of 25-40 percent is directly cited from published DEXA substudies. The directional claim that adequate protein and resistance training *substantially reduce* that share is supported by adjacent (non-GLP-1) intervention trials — a GLP-1-specific intervention trial reporting an exact post-intervention figure is pending. Where this distinction matters, this PDF is explicit about it.

Recent updates.

2026-06-06: Wegovy SA registration status corrected. Wegovy was SAHPRA-registered and launched in South Africa on 18 August 2025. After two Novo Nordisk price cuts (most recent March 2026), the starting dose is approximately R1,873/month and the maximum approximately R3,746/month. Despite registration, Wegovy is generally not covered by SA medical schemes because obesity is not among the 26 Prescribed Minimum Benefits.

Boundary.

Muscle Guard is a self-tracking companion and coach. Not a medical device. Not medical advice. Not affiliated with any prescriber, telehealth provider, or pharmaceutical manufacturer. Always consult your healthcare provider for personal decisions.

BRANCH 1 · WHAT GLP-1s DO TO YOUR BODY

Do you lose muscle on a GLP-1? What the research shows.

By the Muscle Guard research team · Published 2026-06-04 · Last reviewed 2026-06-06 · 12-minute read

Without specific intervention, between roughly a quarter and forty-five percent of weight lost on a GLP-1 receptor agonist can be lean tissue rather than fat — depending on the drug, the dose, and the trial.^{[1][5][4]} With adequate protein intake (1.2 g/kg/day or above) and a resistance-training stimulus, that share drops substantially.^{[8][9]}

The headline number, qualified.

STEP-1 (semaglutide 2.4 mg) produced a mean body-weight reduction of approximately 15 percent. The DEXA substudy reported lean-body-mass reduction of approximately 13 percent — yielding a lean-mass fraction of total weight loss in the region of 40-45 percent.^{[5][2]}

SURMOUNT-1 (tirzepatide 15 mg) produced ~21 percent body-weight reduction. The DEXA substudy reported a 33.9 percent reduction in fat mass and a 10.9 percent reduction in lean mass — i.e. lean mass accounted for roughly a quarter of total weight loss.^{[4][6]}

The Neeland et al. (2024) systematic review consolidated body-composition data across the GLP-1 RA class and concluded that lean-mass-loss share varies meaningfully with molecule, dose, weight-loss magnitude, baseline body composition, and any concurrent intervention.^[1]

Why lean mass loss happens.

Two mechanisms drive lean-mass loss:

Calorie and protein deficit. GLP-1 RAs suppress appetite so effectively that most users sustain a meaningful daily energy deficit. Without enough dietary protein, muscle protein synthesis cannot keep pace with breakdown.

Reduced training stimulus. Many users feel fatigued or queasy in titration weeks and train less. The 'use it or lose it' principle is unforgiving.^[9]

What changes the figure.

Protein at or above 1.2 g/kg/day. Nunes et al. (2022) pooled meta-analysis: intakes above 1.05 g/kg supported up to 1.21 kg additional fat-free mass retention vs lower intakes.^[8] Longland et al. (2016) compared 1.2 vs 2.4 g/kg/day during a 40 percent deficit with training; the higher-protein arm preserved (and slightly increased) lean mass.^[10]

Resistance training during the deficit. Murphy and Koehler (2022): an energy deficit of ~500 kcal/day blunts but does not eliminate lean-mass gains from resistance training; strength gains persist regardless of deficit.^[11] The 2025 Frontiers review reported the majority of resistance-training participants in pooled weight-loss trials maintained or gained lean mass.^[13]

About the magnitude of the drop.

A direct GLP-1 + protein + resistance-training intervention trial reporting the resulting lean-mass-loss share has not yet appeared in the published literature as of June 2026. The directional claim above is based on adjacent intervention trials. When such a trial publishes, the directional language will be replaced with the

cited figure.

What Muscle Guard tracks for you.

Protein adherence — log meals; see daily intake vs target.

Resistance training frequency and volume — log sessions, sets, weight.

Body composition trend — waist, hip, body-fat estimate, visceral-fat rating.

Muscle Guard Score — composite metric capturing fat-loss-without-muscle-loss.

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BRANCH 1 · WHAT GLP-1s DO TO YOUR BODY

GLP-1 side effects, week by week.

By the Muscle Guard research team · Published 2026-06-04 · Last reviewed 2026-06-06 · 9-minute read

Approximately 40-70 percent of users on semaglutide or tirzepatide experience gastrointestinal side effects at some point, most commonly during titration steps.^{[19][18]}

Weeks 1-4 — starting dose.

Standard semaglutide starting dose is 0.25 mg weekly. Tirzepatide is 2.5 mg weekly. Most users report mild to moderate nausea, fatigue, appetite suppression by week 2, and constipation.^[18]

Weeks 5-12 — first titration steps.

Standard semaglutide titration: 0.25 → 0.5 → 1.0 mg over weeks 5-9. Tirzepatide titrates over 16-20 weeks.^{[2][4]}

Weeks 13-24 — approaching peak dose.

Semaglutide approaches 1.7 → 2.4 mg; tirzepatide reaches 12.5-15 mg.^[23]

GI symptoms generally easier than during escalation.

Plateaus possible — weight loss is rarely linear.

Muscle loss becomes visible if not actively prevented.

When to call your healthcare provider.

Severe abdominal pain, especially if persistent — possible pancreatitis.

Vomiting that prevents keeping liquids down.

Signs of dehydration.

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BRANCH 1 · WHAT GLP-1s DO TO YOUR BODY

Mounjaro vs Ozempic vs Wegovy: a 2026 evidence comparison.

By the Muscle Guard research team · Published 2026-06-04 · Last reviewed 2026-06-06 · 10-minute read

In head-to-head trials, tirzepatide produces larger average weight loss than semaglutide at maximal doses: ~21 percent vs ~15-17 percent over 68-72 weeks.^{[4][2][7]} Side-effect profiles are broadly similar; cost and regional access differ substantially.

At a glance.

Ozempic and Wegovy are both semaglutide. **Mounjaro and Zepbound** are both tirzepatide. Ozempic and Mounjaro are registered for type 2 diabetes; Wegovy and Zepbound for chronic weight management.

Efficacy.

Wegovy 2.4 mg vs placebo: 14.9 percent body weight loss at week 68 (STEP-1).^[2]

Tirzepatide 15 mg vs placebo: 20.9 percent body weight loss at week 72 (SURMOUNT-1).^[4]

Head-to-head in T2D (SURPASS-2): tirzepatide produced larger loss than semaglutide.^[7]

Body composition.

DEXA substudies show meaningful lean-mass loss in absence of intervention.^{[5][4]} The Neeland 2024 review notes the absolute lean-mass loss is larger with tirzepatide because total weight loss is larger.^[1]

Cost and access (2026, regional).

South Africa. Ozempic is SAHPRA-registered for type 2 diabetes only and frequently prescribed off-label for weight loss (R2,700-R3,100/month). **Wegovy is now SAHPRA-registered for chronic weight management, launched 18 August 2025** — the first African market for the drug.^{[27][29]} After two Novo Nordisk price cuts (most recent March 2026), the starting dose costs approximately R1,873/month and the maximum approximately R3,746/month.^[28] Mounjaro is registered for type 2 diabetes only, distributed locally by Aspen Pharmacare (R3,000-R6,000/month). Compounded preparations remain widely used at R1,000-R2,500/month. Despite Wegovy's registration, medical schemes generally do not cover GLP-1s for weight loss because obesity is not among the 26 Prescribed Minimum Benefits.^[22]

United States. Wegovy and Zepbound FDA-approved for weight management. Insurance coverage variable.

European Union. Wegovy available in most member states.

Muscle preservation.

None of the drugs preserve muscle by themselves. The lifestyle intervention — protein at 1.2-1.6 g/kg/day and resistance training 2-4 times per week — is the variable that substantially reduces the lean-mass-loss share.

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BRANCH 1 · WHAT GLP-1s DO TO YOUR BODY

Compounded semaglutide and tirzepatide: what the evidence says.

By the Muscle Guard research team · Published 2026-06-04 · Last reviewed 2026-06-06 · 10-minute read

The FDA has flagged significant safety concerns including dosing errors, sterility, and adverse events as of mid-2024 through 2025.^{[14][15]} In South Africa, where branded products historically required out-of-pocket payment, compounded preparations from registered SAPC pharmacies remain in use even after Wegovy's SAHPRA registration.

What the FDA has flagged.

Dosing errors. July 2024: patient inexperience with measuring and self-administering led to overdoses of 5 to 20 times the intended dose.^[14]

Adverse events. Dec 2024: FDA AERS recorded 900+ cases including 17 deaths.^[15]

Sterility concerns from some compounding facilities.

Quality variation. Compounded semaglutide has no FDA approval; profiles are not equivalent to the branded product.

The South African context.

Compounded preparations remain in use at R1,000-R2,500/month, particularly at lower dose strengths where the price gap to branded products is most meaningful. Wegovy's SAHPRA registration and price cuts have narrowed the gap somewhat but not eliminated it.^{[22][27]}

How to assess a compounding pharmacy.

SAPC registration.

API source disclosure and certificate of analysis.

Formulation transparency.

Refrigerated transport and clear storage instructions.

Pharmacist accessibility.

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BRANCH 1 · WHAT GLP-1s DO TO YOUR BODY

Rybelsus and oral semaglutide: how it works, dosing rules.

By the Muscle Guard research team · Published 2026-06-04 · Last reviewed 2026-06-06 · 8-minute read

Rybelsus is oral semaglutide co-formulated with absorption enhancer SNAC.^[17] Bioavailability is approximately 0.4-1 percent; the 30-minute pre-meal fasting rule is non-negotiable.^[16]

What Rybelsus actually is.

Oral semaglutide co-formulated with SNAC, an absorption enhancer that facilitates semaglutide crossing the gastric epithelium.^[17]

Dosing schedule.

Weeks 1-4: 3 mg daily.

Weeks 5-8: 7 mg daily.

Week 9+: 14 mg daily.

The 30-minute fasting window.

Take on an empty stomach with no more than 120 mL of water. Then nothing — no food, drink, or other oral medications — for 30 minutes.^[17]

Side effects.

Mirrors injected semaglutide.^[16]

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BRANCH 2 · HOW TO PROTECT YOUR MUSCLE

The protein target: 1.2 to 1.6 g/kg per day, in plain English.

By the Muscle Guard research team · Published 2026-06-04 · Last reviewed 2026-06-06 · 9-minute read

For adults on a GLP-1, the evidence supports a daily protein target of 1.2-1.6 grams per kilogram of body weight.^{[8][21]} Longland et al. (2016) showed that the upper end of this range, combined with resistance training, preserves and even slightly increases lean mass during a substantial energy deficit.^[10]

Why higher on a GLP-1.

The 0.8 g/kg/day RDA was set for sedentary, weight-stable adults. People on a GLP-1 are neither.^{[21][9]}

Translating the range.

60 kg adult — 72 to 96 g protein per day.

75 kg adult — 90 to 120 g protein per day.

90 kg adult — 108 to 144 g protein per day.

Per-meal target.

Spread across three or four meals containing 25-40 g each.

Practical sources.

2 eggs — 12 g · 100 g chicken breast — 31 g · 100 g salmon — 25 g

200 g Greek yoghurt — 20 g · 1 scoop whey — 24 g · 100 g cottage cheese — 11 g

100 g lentils — 9 g · 100 g firm tofu — 13 g · 1 tin tuna — 25 g

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BRANCH 2 · HOW TO PROTECT YOUR MUSCLE

Resistance training on a GLP-1: the minimum effective dose.

By the Muscle Guard research team · Published 2026-06-04 · Last reviewed 2026-06-06 · 8-minute read

Two to four resistance sessions per week, at RPE 7 to 9, with progressive overload. The minimum effective dose the published intervention arms used to substantially reduce lean-mass-loss share below the 25-40 percent baseline.^{[11][13]}

Why resistance training, specifically.

Aerobic exercise does not provide the muscle-preservation signal at meaningful intensity. Resistance training does.

The minimum effective dose.

Frequency: 2-4 sessions per week.

Volume: 10-20 hard sets per muscle group per week.

Intensity: RPE 7-9.

Progression: add weight, reps, or sets every one to two weeks.

Schoenfeld et al. (2019): 2x/week per muscle group sufficient when weekly volume matched.^[12]

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BRANCH 2 · HOW TO PROTECT YOUR MUSCLE

Body recomposition on a GLP-1.

By the Muscle Guard research team · Published 2026-06-04 · Last reviewed 2026-06-06 · 8-minute read

True recomposition is harder in a sustained calorie deficit, but not impossible.^{[10][11]}

What recomp means here.

For typical GLP-1 users, data supports 'minimise muscle loss' more than 'gain muscle' — and the share of weight lost from lean mass drops substantially below the un-intervened baseline.^{[10][1]}

Tracking cadence.

Weight: weekly.

Waist circumference: weekly.

DEXA or BIA: every 12 weeks while losing.

Progress photos: monthly.

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BRANCH 2 · HOW TO PROTECT YOUR MUSCLE

Muscle drain and 'Ozempic face': what's actually happening.

By the Muscle Guard research team · Published 2026-06-04 · Last reviewed 2026-06-06 · 7-minute read

User-coined terms for losing lean mass alongside fat. Mechanism: sustained calorie and protein deficit without resistance-training stimulus.^{[1][9]} Both are largely preventable.

What it actually describes.

Without intervention, between roughly a quarter and 45 percent of weight lost can be lean tissue.^{[1][5][4]}

The prevention plan.

Protein 1.2-1.6 g/kg/day.

Resistance training 2-4 sessions/week.

Don't push the deficit beyond 0.5-1.0% body weight per week.

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BRANCH 2 · HOW TO PROTECT YOUR MUSCLE

Maintenance after a GLP-1.

By the Muscle Guard research team · Published 2026-06-04 · Last reviewed 2026-06-06 · 8-minute read

STEP-1 extension: participants regained roughly two-thirds of the lost weight within 12 months of semaglutide discontinuation.^[3]

Regain patterns.

Full discontinuation, no plan: 65-80% regained.^[3]

Discontinuation with protein + training continued: smaller and more body-composition-neutral.

Taper to low maintenance dose: smallest regain, with prescriber agreement.

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BRANCH 2 · HOW TO PROTECT YOUR MUSCLE

The Muscle Guard Score, explained.

By the Muscle Guard research team · Published 2026-06-04 · Last reviewed 2026-06-06 · 7-minute read

One number, 0-100, that captures fat-loss-without-muscle-loss. Combines protein (35%), training (25%), weight (20%), body composition (20%).

Why one number, not five.

Humans aren't good at composite mental scoring; we anchor on the metric that moved most this week and miss the one that quietly drifted.

Score bands.

0-49: needs attention.

50-69: okay.

70-84: on track.

85+: excellent.

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BRANCH 3 · THE RESEARCH, UPDATED

GLP-1s in South Africa, 2026: cost, access, the compounded landscape.

By the Muscle Guard research team · Published 2026-06-04 · Last reviewed 2026-06-06 · 8-minute read

Ozempic SAHPRA-registered for T2D at R2,700-R3,100/month. **Wegovy SAHPRA-registered for chronic weight management since 18 August 2025** at R1,873-R3,746/month after two price cuts.^{[27][29]} Compounded semaglutide and tirzepatide from SAPC pharmacies remain at R1,000-R2,500/month.^[22]

What Ozempic actually costs.

Ozempic 0.25 mg (starter): R1,800-R2,200/month.

Ozempic 0.5 mg: R2,300-R2,700/month.

Ozempic 1.0 mg: R2,700-R3,100/month.

Compounded semaglutide: R1,000-R2,500/month.

Wegovy — registered since August 2025.

Novo Nordisk launched Wegovy in South Africa on 18 August 2025 as the first African market. SAHPRA-registered as Schedule 4 for chronic weight management.^[29] Available in five dose strengths.^[27]

Two price cuts since launch — most recent March 2026 — in response to Mounjaro competition. Starting dose ~R1,873/month (down from ~R3,090); maximum dose ~R3,746/month (down 27 percent).^[28]

Medical aid coverage.

Despite SAHPRA registration, Wegovy is generally not covered by SA medical schemes because obesity is not currently among the 26 Prescribed Minimum Benefits.^[22]

For diabetes (registered indication): covered with chronic-condition motivation.

For weight loss alone: almost universally not covered.

For weight loss with comorbidity: occasionally approved with BMI > 35 plus documented comorbidities.

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BRANCH 3 · THE RESEARCH, UPDATED

Getting the most from your GLP-1 doctor visit.

By the Muscle Guard research team · Published 2026-06-04 · Last reviewed 2026-06-06 · 6-minute read

Most consults are 15-20 minutes. The productive ones start with a one-page summary your prescriber reads in 60 seconds.

The five-day prep window.

Day -5: Run a draft Doctor PDF.

Day -1: Write three questions.

Three questions to ask.

Is the dose right for the next 4-6 weeks?

Which side effect should I track most carefully?

What would make you change the plan?

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BRANCH 3 · THE RESEARCH, UPDATED

Privacy and GLP-1 apps.

By the Muscle Guard research team · Published 2026-06-04 · Last reviewed 2026-06-06 · 7-minute read

Health data is 'special category' under GDPR Article 9 — requiring explicit consent.

Your rights.

Access, rectification, erasure, portability, restriction, objection.

How Muscle Guard handles privacy.

EU-region data hosting for EU users.

No third-party advertising. No data sale.

Deletion in one tap.

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Citations.

All sources used across this library. Click any DOI/URL to open the original.

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Methods note.

Sources are peer-reviewed unless otherwise labelled. Regulatory body publications (FDA, SAHPRA) and journalism sources (Spotlight SA, Reuters, eHealth Magazine) are used for market and access claims, labelled inline. The 25-40 percent un-intervened lean-mass-loss share is directly cited. The 'drops substantially' phrasing for post-intervention is supported by adjacent intervention trials; a GLP-1-specific trial reporting an exact figure is pending and will be cited here when it lands.

Errors.

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